



Casting a shadow: Productivity of formal firms and informality

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Abstract

Informality is a salient feature of most developing economies. Two important concerns arise from informality. First, informal firms tend to be less productive than formal firms. Second, the competition from informal firms can weigh on the productivity of firms in the formal sector. These challenges can lower the aggregate productivity and thereby retard overall development in countries where the informal sector is pervasive. Our analysis provides robust evidence on both issues using firm-level survey data for a large cross section of countries. We also examine the extent to which firm characteristics and business environment quality shape the productivity gap between formal and informal firms as well as the impact of informal competition on the productivity of formal sector firms. The results show that, on average, the labor productivity of informal firms is about one-fourth that of formal firms. Moreover, competition from informal firms lowers the labor productivity of formal firms by 20%–24%. In line with the “parasite” view of informality, the negative impact of informal competition on the productivity of the formal sector firms is more pronounced when the business environment is less conducive to operating formally due to high corruption, burdensome regulations, and weak institutions.

KEYWORDS

business climate, formal firm, informal competition, productivity gap

JEL CLASSIFICATION

D22; E26; O17

1 | INTRODUCTION

Understanding informality and its economic implications is of paramount importance for poverty-reducing and welfare-enhancing policies. Informal businesses absorb nearly 70% of the labor force in emerging markets and developing economies (ILO, 2018). Despite its potential contribution to the economy, there is a concern that the informal sector undermines aggregate productivity. This can happen for at least two reasons. First, informal firms are highly inefficient vis-à-vis formal sector firms, which lowers aggregate productivity. Second, the competition from informal firms can weigh on the productivity of firms in the formal sector.

This paper uses firm-level survey data collected by the World Bank to analyze the aforementioned concerns of informality. Specifically, we estimate the difference in labor productivity between formal and informal firms (henceforth, productivity gap) and the impact of competition from informal firms (henceforth, informal competition) on the labor productivity of the formal firms. We contribute to the literature in the following ways.

First, we provide new evidence on the labor productivity gap and the impact of informal competition (on formal firms' labor productivity) using firm-level survey data for a large cross section of mostly developing countries.

Second, we pay due attention to potential endogeneity concerns in estimating the impact of informal competition. To this end, we proxy the informal competition faced by a given formal firm by the level of informal competition experienced by all other formal firms in the same location–size–industry category. This strategy helps substantially mitigate endogeneity concerns. Additional endogeneity checks are undertaken by testing some implications of the “parasite” view of informality. These endogeneity checks are motivated by the study of Distinguin, Rugemintwari, and Tacneng (2016), among others, and control for the heterogeneous effects of informal competition on the productivity of the formal firms. For the productivity gap analysis, our approach is more modest due to data limitations.

Third, the World Bank's Enterprise Surveys (ES) we use are conducted using a harmonized sampling methodology and questionnaire that facilitate cross-country comparability while allowing to control for heterogeneity across data points within a country. Previous studies (reviewed in Section 2) using microdata are largely restricted to a single country or a narrow set of countries, and it is not clear if the results of these studies hold more generally.

Fourth, we explore the heterogeneity in the size of the productivity gap and the impact of informal competition. As discussed in detail later in this paper, informality may impact countries and firms differently. For instance, informal firms operating at the fringes of the formal–informal divide are likely to enjoy a greater “unfair” advantage over formal firms when the cost of operating formally versus informally is higher and the benefits lower. This is more likely to be the case when the regulatory burden on formal firms is greater, corruption is higher, and enforcement of rules and laws is weaker. Similarly, the productivity gap may vary if, for example, informal firms are less capable of benefiting than the formal firms from factors that boost productivity such as managerial experience and location. Exploring heterogeneity is important from both academic and policy-making standpoints. One reason is that the size of the informal sector has remained large in many countries despite the economic progress and several formalization programs. The heterogeneity analysis allows us to identify which factors exacerbate the harmful effects of informality and thereby helps design more targeted policy responses.

Our results show that the labor productivity of informal firms is on average about one-fourth that of formal firms (Figure 1). Firm characteristics such as higher age and location in the capital city that tend to boost labor productivity are found to benefit the formal firms more than the informal firms.

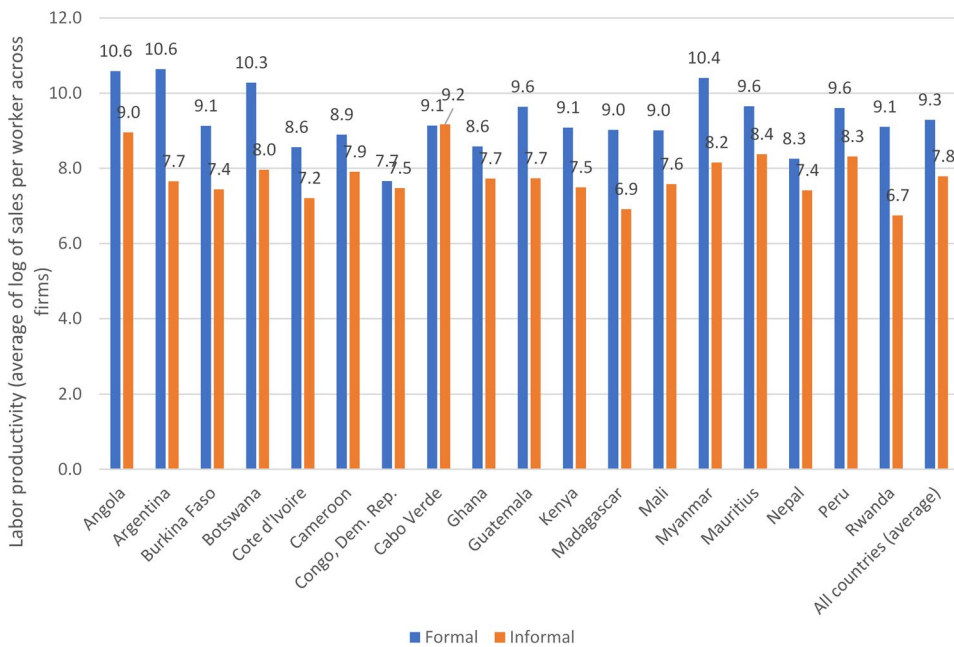


FIGURE 1 Productivity gap between formal and informal firms. *Source:* Authors' own calculations based on data from Enterprise surveys

Thus, the productivity gap rises with these characteristics. The broader implication is that informal firms are less capable of benefitting from productivity-enhancing factors than formal firms.

Moreover, we find that the labor productivity of formal firms facing informal competition is on average about 76%–80% of the labor productivity of formal firms that do not experience informal competition (Figures 2 and 3). Further, consistent with the “parasite” view, this adverse effect of the informal competition is stronger in countries where the business environment is less conducive to operating formally due to higher corruption, more burdensome regulations, and weaker enforcement of rules and laws.

Due caution is necessary for interpreting our results causally. We present correlations or associations that suggest possible underlying causal links. These associations do not establish definitive causal relationships. Nonetheless, we conduct many checks against endogeneity concerns, especially for the informal competition results, as detailed data permit a more rigorous robustness analysis.

2 | LITERATURE REVIEW

2.1 | Productivity difference between formal and informal firms

The productivity gap between formal and informal firms has been discussed in several papers (Benjamin & Mbaye, 2012; Farrell, 2004; Gelb, Mengistae, Ramachandran, & Shah, 2009; La Porta & Shleifer, 2008, 2014; Perry et al., 2007). As pointed out by Farrell (2004, p. 30), firms in the informal sector tend to “become trapped in a self-reinforcing dynamic that confines them to subscale, inefficient, low-productivity work,” leading to a productivity gap across the formal–informal divide. This may happen due to important differences between informal and formal firms such as unequal access to public services (Steel & Snodgrass, 2008), difference in the business environment quality (Gelb et al.,

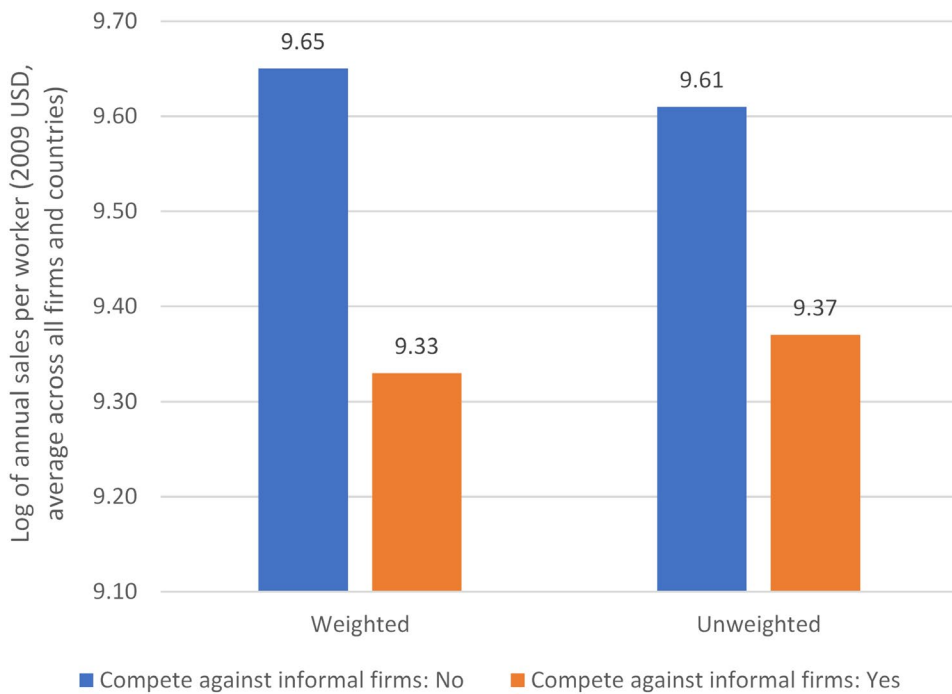


FIGURE 2 Difference in labor productivity between firms that compete against informal firms and those that do not. *Source:* Authors' own calculations based on Enterprise surveys data for 110 countries. *Note:* The figures shown are computed by taking the average value of the (log of) labor productivity for all firms in the sample that compete against informal firms and those that do not. Results are shown with and without using the sampling weights

2009), difference in the nature of informality (La Porta & Shleifer, 2008), difference in the motivation (subsistence vs. entrepreneurial) to start the business (Perry et al., 2007), and difference in the ability to grow (De Mel, McKenzie, & Woodruff, 2010).

A range of productivity gap estimates are found in the literature, partly because studies use various samples and different definitions of informality. Using a sample of firms from seven Latin American and Caribbean countries, Perry et al. (2007) find that the labor productivity of firms that started their operations without being formally registered (referred to as informal firms) is about 29% lower than the productivity of firms that have always operated formally. La Porta and Shleifer (2008) use, like us, the World Bank's ES data for a cross section of developing countries. They find that in the median sample country, informal (unregistered) firms have only 15% of the value added per worker of formal (registered) firms. The authors suggest that the observed gap stems from the smaller size of the informal firms compared to formal firms and differences in input use, especially management quality. Benjamin and Mbaye (2012) focus on the three capital cities of Benin, Burkina Faso, and Senegal. The authors use a multifaceted definition of informality based on the business' activity size, registration status, bank credit availability, workplace mobility, and financial reporting. They confirm that labor productivity is much lower among informal firms than the formal firms, largely due to smaller-sized informal firms in the sample. Echevin and Murtin (2009) use matched employer–employee data collected by World Bank and Afristat in 2003 for Senegal. They corroborate that labor productivity is much lower for informal (unregistered) firms than the formal (registered) firms. Surveying existing studies around the world, Farrell (2004) notes that informal businesses operate at just half the average productivity level of formal companies in the same sector.¹

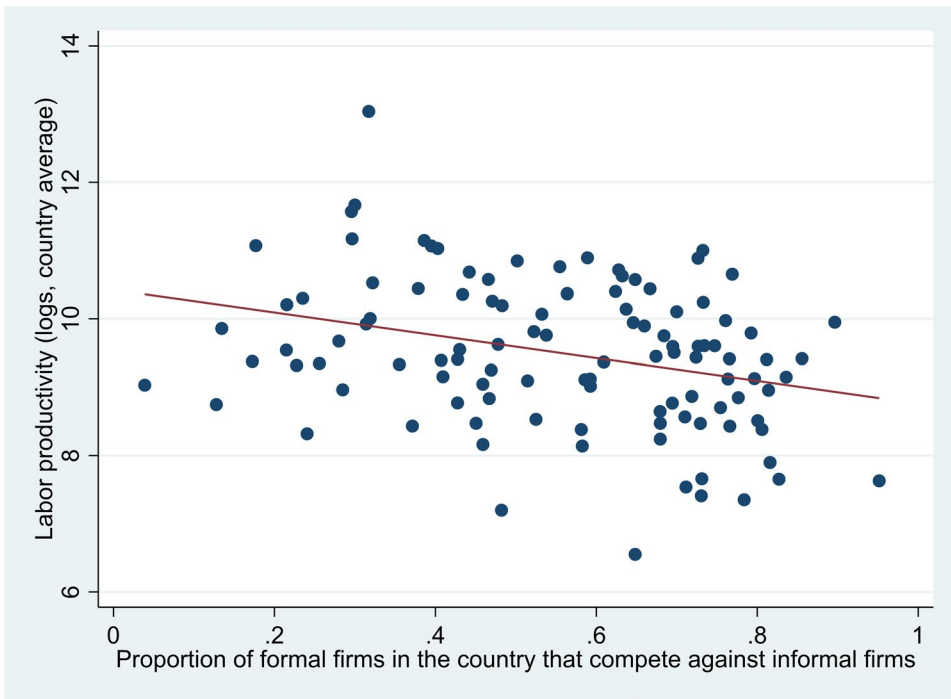


FIGURE 3 Relationship between labor productivity and proportion of firms in the country that face competition from informal sector firms. *Source:* Authors' own calculations based on Enterprise surveys data for 110 countries

2.2 | Impact of informal competition on the productivity of formal sector firms

Another strand of the literature investigates how competition from informal firms affects the productivity and performance of firms in the formal sector. The two prominent perspectives are the “parasite” view and the “dual economy” view (La Porta & Shleifer, 2008, 2014). The “dual economy” view asserts that formal and informal firms are fundamentally different. Informal labor is a by-product of poverty and failure to find good jobs in the formal sector. Informal businesses are small, inefficient, of subsistence nature, and their productivity is too low to allow them to compete and survive in the formal sector. Thus, there is no spillover from informal to formal sector (La Porta & Shleifer, 2008, 2014; Rothenberg et al., 2016).

In contrast, the “parasite” view considers informal firms as capable of competing against formal firms. Informality arises from a rational decision of firms to avoid costly regulations and taxes—the *rational exit model* (De Mel, McKenzie, & Woodruff, 2011; Maloney, 2004). Such informal firms operate on the fringes of the formal–informal divide, providing stiff competition to the formal sector firms. In general, on a level-playing field, competition can lead to higher productivity through industry rationalization and improved efficiency (Djankov & Murrell, 2002). However, informal firms do not pay taxes and do not comply with costly regulations, whereas formal firms must shoulder the cost of compliance. This “unfair” advantage allows informal firms to charge lower prices and “steal” market share from the formal firms. Thus, formal firms’ production declines with consequent loss of scale-related economies. Lower prices impinge on the profitability of the formal firms, reducing their

incentive and restricting their access to critical funds for investment and innovation. Thus, informal competition can adversely affect the productivity of formal firms.

The empirical evidence on the impact of informal competition on the productivity of formal firms is rare. In an early attempt, Tokman (1978) finds that in the city of Santiago, informal foodstuff commercial establishments can successfully compete with formal sector counterparts (modern supermarkets). However, the effect of informal competition on the productivity of formal establishments is not discussed. Gonzalez and Lamanna (2007) analyze firm-level survey data on formal manufacturing firms collected by the World Bank's ES for 14 countries in Latin America. Their findings suggest that about 40% of the firms in the region face significant informal competition, with sizeable variation across industries and firm sizes. However, the study does not offer any direct evidence on the impact of informal competition on formal firms' productivity. Using ES data for Nicaragua, Pisani (2015) explores the firm characteristics that determine the likelihood of formal firms to face informal competition. Mendi and Costamagna (2017) use data from ES to estimate the impact of informal competition on formal sector firms' innovation. Again, these studies do not assess the direct impact of informal competition on formal firms' productivity.

La Porta and Shleifer (2008) use ES data to examine the expected effects of informality on the formal sector firms. However, the authors do not investigate the impact on productivity. Instead, they analyze surveyed formal firms' responses to whether "anticompetitive and informal" practices are an obstacle for their business. The authors acknowledge that the question is noisy because anticompetitive practices can result from the behavior of the formal sector firms. Firms' view of informality as an obstacle may also be driven by the perceived impact of the informal activity on total revenues or profits rather than on productivity. Moreover, La Porta and Shleifer (2008) use old data collected between 2002 and 2007.

Other studies highlight the negative effects of informality on the formal sector and the overall economic development. Perry et al. (2007) argue that informality can have negative effects on formal firms' investment and innovation decisions because it reduces their market share and profitability. However, they do not provide any empirical evidence to support this claim. Roza and Winkler (2019) find that, in Colombia, an increase in internally displaced persons expands the informal sector, which in turn adversely affects the total output of the formal sector firms. Loayza (1996) suggests that a large informal economy may lower tax revenues and increase public service congestion. The negative effects of congestion, however, fall on all firms and therefore cannot be linked to the informal competition.

3 | DATA DESCRIPTION AND MAIN VARIABLES

3.1 | Data and empirical strategy

We define informality as the lack of formal registration of firms. Simply put, unregistered firms belong to the informal sector, whereas formal firms are those that are registered with the relevant authorities. The merits and limitations of various definitions of informality are discussed in Benjamin and Mbaye (2012).

Our main data source is firm-level survey data collected by the World Bank's ES on formal or registered firms with five or more employees. These surveys are representative of the nonagricultural and nonfinancial registered private economy—excluding firms with fewer than five employees. A common sampling methodology—stratified random sampling—and a harmonized questionnaire are used in all surveyed countries. These data are complemented with data on informal businesses

collected by World Bank surveys in a few business centers (cities) in some countries. Based on local information, World Bank contractors identified regions or neighborhoods in the target city with high levels of informal activity. These neighborhoods were then divided into enumeration blocks, which were then surveyed on foot. This informality survey is a pure random sample of informal firms in the city. Note that the sample is not necessarily representative of the informal sector in the city due to lack of a proper sampling frame. Thus, due caution is required in interpreting our results. We share this problem with the literature (see, e.g., La Porta & Shleifer, 2008). The firm-level data on formal and informal firms used in our analysis are available online.²

The country coverage of the sample used in the empirical analysis varies. The harmonized informality surveys used for the productivity gap analysis are available for 18 countries. The sample includes 3,145 informal firms and 6,591 formal firms. The informal surveys were carried out between 2009 and 2014, whereas data on formal firms were collected between 2007 and 2012. In each country, we use the latest round of data available. Table 1 provides the list of countries along with the sample size of informal and formal firms.

To analyze the impact of informal competition on the labor productivity of the formal sector firms, we use ES data only for the formal sector firms. The sample here consists of 42,742 firms located in 110 countries, mostly emerging markets and developing economies. These surveys were conducted between 2008 and 2016. We use the latest round of ES available in each country. Table A1 in the online Appendix provides the sample distribution by country.

Sampling weights (normalized to unity for each country) are available only for formal ES because they are representative and are used throughout the analysis. For informal firm surveys, sampling

TABLE 1 Sample of countries for formal vs. informal labor productivity comparison

Country	Number of informal firms	Number of formal firms
Angola	65	158
Argentina	134	908
Botswana	76	225
Burkina Faso	80	323
Cabo Verde	53	120
Cameroon	112	332
Congo, Dem. Rep.	318	438
Cote d'Ivoire	61	142
Ghana	537	504
Guatemala	204	421
Kenya	369	600
Madagascar	121	165
Mali	62	174
Mauritius	75	148
Myanmar	208	521
Nepal	108	343
Peru	232	891
Rwanda	330	178
All countries	3,145	6,591

weights are not provided. Therefore, we assign a weight of 1 to each informal firm and normalize it so that the weights sum to 1 for each country.

3.2 | Dependent variable

The dependent variable in all our regressions is firm-level labor productivity. It is defined as (log of) the total sales of a firm (deflated and expressed in 2009 USD) during the last fiscal year divided by the number of workers in the firm.³ For the first sample of 18 countries used for estimating the productivity gap, the average value of labor productivity (in logs) equals 8.5 and the standard deviation is 1.8. For the second sample of 110 countries used for estimating the impact of informal competition, the mean value of labor productivity (in logs) equals 9.47 with a standard deviation of 1.85.

3.3 | Main explanatory variables

For the productivity gap analysis, the main explanatory variable is a dummy that equals 1 if the firm is informal and 0 otherwise (informal firm).

For the impact of informal competition on the labor productivity of the formal firms, the main explanatory variable is a measure of informal competition faced by formal firms. Specifically, the ES ask formal firms whether they compete against informal firms. The variable cannot be used directly in the regressions as it is likely to be endogenous to many firm characteristics. In fact, whether a firm faces informal competition may itself depend on its level of labor productivity (reverse causality problem).

One solution suggested in the literature is to proxy the informal competition faced by a formal firm with the average level of informal competition faced by all other formal firms (except the firm in question) in the same location, industry, and size group (henceforth, cell). Reverse causality from a given formal firm's productivity to the informal competition experienced by other formal firms in the cell is highly unlikely although the possibility cannot be ruled out completely (see, e.g., Aterido et al., 2011). Moreover, since the cell excludes the firm in question, the cell average of informal competition is unlikely to be correlated with the characteristics of the firm in question. This helps to mitigate the omitted variable bias problem. Using the cell average also helps control for potential measurement errors if some firms choose not to respond or misstate the informal competition that they experience (Pounov, 2016). A few examples of studies that use cell averages in the sense described previously include Dollar, Hallward-Driemeier, and Mengistae (2006), Fisman and Svensson (2007), De Rosa, Gooroochurn, and Görg (2010), Aterido, Hallward-Driemeier, and Pages (2011), and Amin and Soh (2020).

In constructing the cells, a choice needs to be made about location, industry, and size. Cells defined at a more disaggregated level are more likely to contain firms that face similar levels of informal competition. However, the number of firms in each cell is reduced, raising concerns about the reliability of the cell average. In line with the literature, we define the level of informal competition faced by a formal firm as the proportion of all firms in the region–industry–size cell excluding the firm in question that reports competing against informal firms. For firm size, we use the ES classification of small (fewer than 20 employees), medium (20–99 employees), and large (100 or more employees). For region, we use the subnational regions that match the ES strata. For industry, we use the three-digit ISIC Rev. 3.1 groupings. Excluding cells with fewer than three firms, we are left with 7,081 region–industry–size cells.⁴

To further address endogeneity concerns, we control for several characteristics that could be potentially correlated with both labor productivity and our main explanatory variable (discussed later). In

Section 5.1, we discuss in detail another test that we use against endogeneity concerns about the relationship between informal competition and the labor productivity of the formal firms. This is based on the idea that the “unfair” advantage enjoyed by the informal firms over formal firms is greater when the benefits of operating formally are lower and costs higher.

4 | PRODUCTIVITY GAP BETWEEN FORMAL AND INFORMAL FIRMS

This section explores whether informal firms are less productive compared to the formal sector firms—the first of the two ways in which informality can undermine aggregate productivity in a country.

4.1 | Empirical specification and controls

The empirical estimation of the productivity gap relies on the following equation:

$$Y_i = \alpha + \beta X_i + \sum_{k=1}^K \gamma_k Z_{ik} + \sum_{k=1}^K \delta_k X_i Z_{ik} + CFE_i + \varepsilon_i, \quad (1)$$

where the subscript i denotes the firm, Y is the dependent variable (labor productivity), X is the main explanatory variable (informal firm dummy), $\{Z_k\}_{k=1}^K$ is a set of firm-level controls, XZ_k are interaction terms that aim to capture how the productivity gap may vary depending on basic firm characteristics such as firm’s age and number of workers (details below), CFE is country fixed effects, and ε is the error term. The equation is estimated using the ordinary least squares (OLS) method with Huber–White robust standard errors.

We use a parsimonious structure of controls due to data limitations and limited research on the drivers of productivity in the informal sector.⁵ Our choice of controls is guided by existing studies on the determinants of labor productivity largely focused on formal firms.

Our first set of controls includes dummy variables that indicate the country where the firm is located (country fixed effects). Thus, differences in overall economic development, financial development, and quality of macro-level institutions are purged from our estimation results.

Next, we control for the number of workers (logs) employed at the firm. More workers imply larger firms that can lead to higher productivity due to the presence of fixed costs and greater allocative efficiency (Bartelsman, Haltiwanger, & Scarpetta, 2013; Tybout, 2000). However, diminishing returns to labor may prevail as fixed capital and resources are spread out over a larger number of workers. In that case, labor productivity may decline as the number of workers increases. Amin and Islam (2015) use ES data on informal firms and find that labor productivity among informal firms declines sharply with firm size.

The link between productivity and age of the firm has been extensively investigated. Age-related effects among surviving firms can be explained by scale economies gained from expansion over time, vintage effects due to younger firms employing new and improved technology or equipment, selection effects that weed out inefficient firms, and learning by doing (see, e.g., Bahk & Gort, 1993; Jensen, Robert, & Stiroh, 2001; Jovanovic, 1982). If formal and informal firms differ systematically in their age, it could induce an omitted variable bias in our estimates. Thus, we control for the (log of) age of the firm.

Third, productivity differences can also arise across industries. If industries that have high productivity also happen to be the ones that have low or high ratios of informal to formal firms, our main

results could suffer from omitted variable bias problems. To guard against this possibility, we control for a dummy variable equal to 1 if the firm is a manufacturing firm and 0 if it belongs to the services sectors. Information on industry at a more disaggregated level is not available for the informal firms.

Next, differences in the quality of management have been found to impact firm productivity (see, e.g., Bloom & Van Reenen, 2007; Syverson, 2011). Differences in education and experience of the top manager are possible drivers of differences in management quality. If the overall quality of the firm's top management varies systematically between formal and informal firms, our estimates could suffer from omitted variable bias. Thus, we control for the (log of) number of years of work experience the top manager of the firm has in the industry (manager experience).

Several studies have shown that female-owned or managed firms tend to be much smaller, less profitable, and less efficient than the male-owned or managed firms (Du Rietz & Henrekson, 2000; Islam, Gaddis, Palacios-Lopez, & Amin, 2018). This could be due to difficulties that women face in obtaining credit and accessing institutions, laws that discriminate against women, and culture that tends to discourage women's work. Thus, we control for a dummy variable equal to 1 if the main decision maker of the firm is a woman and 0 otherwise.

Our last control is for the location of the firm within the country. Capital cities are typically among countries' largest economic centers and, therefore, can offer agglomeration benefits: larger markets, better infrastructure to access markets and operate, a larger pool of workers, and greater technology spillovers (Rosenthal & Strange, 2004). This is likely to impact labor productivity positively. To guard against the implied omitted variable bias problem, we control for a dummy variable equal to 1 if the firm is in the capital city of the country and 0 otherwise.

Information on the capital stock used by the firms is not available in informality surveys. Formal and informal firms were asked if they use machinery and equipment. However, this is too crude a measure of capital stock especially for the formal sector firms. A failure to control for the capital stock may lead to omitted variable bias in our empirical estimation of the productivity gap. Therefore, due caution is necessary for interpreting our results.

4.2 | Base regression results

Our base regression results for the productivity gap are given in Table 2. These results show that irrespective of the controls, labor productivity is lower for informal firms than for formal firms. The productivity differential is quantitatively large ranging between -1.36 and -1.65 (log points). The differential is significant at the 1% level in all the specifications. The most conservative estimate of the productivity gap is obtained in the final specification that includes all the controls. According to this estimate, the labor productivity of informal firms is lower by about 74% compared to a formal firm.

Regarding the various controls, labor productivity is significantly higher (at the 5% or 1% level) for firms with more experienced managers, older firms, firms located in the capital cities, services sector firms versus manufacturing, firms with a male versus female top manager, and the relatively smaller firms.⁶

4.3 | Interaction term results

Regression results for the interaction terms are provided in Table 3. Starting with the final specification given earlier, we add the interaction terms one by one (columns 2–5) and then all at once (column 6). The interaction terms used are the ones between the informal firm dummy and the following

TABLE 2 Base regression results for the labor productivity gap

Dependent variable: labor productivity	(1)	(2)	(3)	(4)	(5)
Informal firm: Y = 1, N = 0	−1.510*** (0.056)	−1.464*** (0.052)	−1.652*** (0.083)	−1.448*** (0.086)	−1.361*** (0.087)
Number of workers (logs, workers)			−0.084*** (0.025)	−0.076*** (0.027)	−0.090*** (0.027)
Firm age (logs)				0.163*** (0.035)	0.113*** (0.045)
Manufacturing: Y = 1, N = 0				−0.395*** (0.055)	−0.421*** (0.054)
Manager experience (years, logs)					0.088** (0.041)
Top manager is a woman: Y = 1, N = 0					−0.189*** (0.064)
Capital city: Y = 1, N = 0					0.224*** (0.060)
Country fixed effects	No	Yes	Yes	Yes	Yes
Constant	9.284*** (0.041)	9.259*** (0.039)	9.515*** (0.090)	9.229*** (0.112)	9.061*** (0.128)
Number of observations	9,736	9,736	9,736	9,736	9,736
R-squared	0.168	0.281	0.283	0.297	0.303

Note: Robust standard errors in parentheses. Significance is denoted by ***1% and **5%.
N = no; Y = yes.

TABLE 3 Interaction term results for the labor productivity gap between formal and informal firms

Dependent variable: labor productivity	(1)	(2)	(3)	(4)	(5)	(6)
Informal firm: Y = 1, N = 0	−1.361*** (0.087)	−0.621*** (0.181)	−1.054*** (0.118)	−1.090*** (0.108)	−1.298*** (0.088)	0.075 (0.192)
Number of workers (logs, workers)	−0.090*** (0.027)	−0.100*** (0.026)	−0.024 (0.029)	−0.085*** (0.026)	−0.086*** (0.027)	−0.060*** (0.029)
Firm age (logs)	0.113** (0.045)	0.283*** (0.052)	0.109** (0.044)	0.105** (0.045)	0.111** (0.044)	0.271*** (0.051)
Manufacturing: Y = 1, N = 0	−0.421*** (0.054)	−0.422*** (0.054)	−0.418*** (0.054)	−0.418*** (0.054)	−0.421*** (0.054)	−0.425*** (0.054)
Manager experience (years, logs)	0.088** (0.041)	0.147*** (0.042)	0.116*** (0.040)	0.097** (0.040)	0.087** (0.041)	0.141*** (0.042)
Top manager is a woman: Y = 1, N = 0	−0.189*** (0.064)					−0.006 (0.116)
Capital city: Y = 1, N = 0	0.224*** (0.060)	0.216*** (0.060)	0.209*** (0.060)	0.516*** (0.085)	0.227*** (0.061)	0.500*** (0.085)
Informal firm*firm age		−0.358*** (0.070)				−0.358*** (0.070)
Informal firm*number of workers			−0.246*** (0.067)			−0.219*** (0.069)
Informal firm*capital city				−0.529*** (0.115)		−0.550*** (0.114)
Informal firm*top manager is a woman					−0.261*** (0.078)	−0.311** (0.139)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Constant	9.061*** (0.128)	8.500*** (0.157)	8.782*** (0.137)	8.822*** (0.126)	9.030*** (0.126)	8.236*** (0.164)
Number of observations	9,736	9,736	9,736	9,736	9,736	9,736
R-squared	0.303	0.307	0.304	0.306	0.304	0.317

Note: Robust standard errors in parentheses. Significance is denoted by ***1% and **5%.

N = no; Y = yes.

firm-level features: age of the firm, number of workers, capital city dummy, and the dummy for the gender of the top manager.

Results in Table 3 show that all the interaction terms are negative, large, and statistically significant at the 5% level in one case and at the 1% level in the remaining cases. This holds when the interaction terms are included individually or all simultaneously. Recall that firm's age and location in the capital city tend to boost labor productivity in the baseline results (see Table 2). Thus, the interaction term results in Table 3 imply that the benefits of being older and being in the capital city are much smaller for informal firms than for formal firms. Also, recall that labor productivity in our sample declines as the number of workers increases, and it is lower for female-managed firms than for male-managed firms. Results in Table 3 show that the decline in labor productivity associated with having a larger number of workers and with having a female versus male manager is much bigger for informal firms than for the formal firms. Thus, compared to formal firms, informal firms tend to benefit less from factors that boost labor productivity and suffer more from factors that hamper labor productivity.

To assess the magnitude of heterogeneities, we consider the results in Table 3 where the interaction terms are included separately (columns 2–5). We provide a couple of examples to illustrate the point. First, for firms located in the capital cities, the labor productivity of informal firms is lower by about 80% than that of formal firms; the corresponding figure for the rest of the country is lower at 66%. For male-managed firms, the labor productivity of the informal firms is lower by about 73% than that of a formal firm; the corresponding figure for female-managed firms is larger at 79%. Thus, there are large variations in the productivity gap depending on the characteristics of the firms.

5 | LABOR PRODUCTIVITY OF THE FORMAL FIRMS THAT COMPETE AGAINST THE INFORMAL FIRMS

In Section 4, we reported evidence of a large labor productivity gap between formal and informal firms. Thus, informality undermines aggregate productivity and retards overall economic development. This section focuses on the second way in which informality lowers aggregate productivity. This happens through the adverse effect of informal competition on the labor productivity of the formal sector firms.

5.1 | Empirical specification and control variables

Recall that regression results for the relationship between the labor productivity and informal competition are based on a sample of 42,742 formal (registered) firms. The overall impact of informal competition on the labor productivity of the formal firms is estimated using the following equation:

$$Y_i = \alpha + \beta X_i + \sum_{k=1}^K \gamma_k Z_{ik} + CFE_i + IFE_i + \varepsilon_i, \quad (2)$$

where the subscript i denotes the firm, Y is the dependent variable (labor productivity as defined earlier), $\{Z_k\}_{k=1}^K$ is a set of firm-level controls, CFE is country fixed effects, IFE is industry fixed effects where industry is defined at the three-digit ISIC Rev. 3.1 level, and ε is the error term. X is the main explanatory variable that is a measure of informal competition faced by the formal firms. As discussed in Section 3.3, this equals the proportion of formal sector firms in the region–industry–size cell (excluding the firm in

question) that report competing against informal sector firms. The equation is estimated using the OLS method with Huber–White robust standard errors.

Since we now focus only on the formal firms, factors such as exporting activity, foreign ownership of firms, and use of capital may be important. Moreover, cross-country comparable data are available for these variables. Thus, the control variables included in Equation 2 are those used for the productivity gap analysis in Section 4 and controls for exports, foreign ownership, and a proxy measure of capital use (described later). Equation 2 also includes industry fixed effects (35 industries defined at the three-digit ISIC Rev. 3.1 level), whereas Equation 1 includes a simple dummy for manufacturing versus the rest due to data limitations.

The motivation for all the controls has already been discussed in Section 3.1 except for exports, foreign ownership, and capital. Exporting activity and foreign ownership have been linked to productivity in several studies although the direction of causality remains an open debate (Melitz, 2003; Wagner, 2007). Therefore, we control for the proportion of firms' annual sales made abroad (exports). For foreign owners, we use a dummy variable equal to 1 if foreigners (individuals or companies) own at least 10% of the firm and 0 otherwise. Information on the actual capital stock (e.g., book value and replacement value of machines, equipment) is missing due to nonresponse for more than half the firms. Therefore, following Amin and Soh (2020) among others, we use a proxy measure of capital use that is a dummy variable equal to 1 if the firm spent on the purchase of fixed assets during the past year and 0 otherwise. We acknowledge that this is a partial measure of capital use and therefore our results could reflect differences in capital use across firms.

The second part of the empirical analysis explores the heterogeneity in the labor productivity and informal competition relationship. This is motivated by the “parasite” view of informality suggesting that firms chose to go informal not because they cannot compete with formal firms but to avoid taxes and other costly regulations. The “unfair” advantage enjoyed by the informal firms hurts the formal sector firms. We argue that this “unfair” advantage is greater when the cost of operating formally versus informally is higher or the benefits lower. Distinguin et al. (2016) make a similar point in their analysis of the impact of informal competition on access to finance for formal SMEs. They find that the impact is negative and more so in countries with higher corruption, higher taxes, and weaker rule of law. For instance, they argue that a strong rule of law raises the likelihood of informal activity detection; therefore, informal firms are forced to keep their operations small, weakening their capacity to affect formal SMEs.

Like Distinguin et al. (2016), we focus on corruption, regulatory burden, and quality of enforcement as factors that exacerbate or mitigate the effects of informal competition. Thus, our testable hypothesis is that the negative impact of informal competition on the labor productivity of the formal firms should be stronger (more negative) in countries where formal firms face higher corruption, greater regulatory burden, and weaker enforcement mechanisms for implementing laws and rules.

The heterogeneity analysis serves two purposes. First, it unveils the factors that exacerbate the impact on informality. This is important for a more targeted policy intervention. Second, the hypothesis stated in the previous paragraph assumes that mechanisms specific to the functioning of the informal sector are at play. This hypothesis is unlikely to hold if our informal competition variable is a mere proxy for other economic development correlates. Thus, testing for the hypothesis serves as a useful check against endogeneity concerns.

The heterogeneity analysis is carried out by adding interaction terms to the final specification in Table 4. We first add the interaction term between the informal competition variable and GDP per capita (PPP adjusted and at constant 2011 international dollars from the World Development Indicators, World Bank) as the latter is a useful proxy measure of the broad business environment. Next, we use interaction terms between the informal competition variable and measures of corruption, regulatory

TABLE 4 Informal competition and labor productivity

Dependent variable: labor productivity	(1)	(2)	(3)	(4)	(5)
Informal competition (cell average)	−0.698*** (0.080)	−0.217*** (0.078)	−0.274*** (0.076)	−0.273*** (0.075)	−0.262*** (0.074)
Number of workers (logs)			−0.164*** (0.018)	−0.187*** (0.018)	−0.247*** (0.019)
Firm's age (logs)				0.186*** (0.028)	0.206*** (0.027)
Top manager experience (logs)				0.075** (0.033)	0.069*** (0.032)
Top manager is a woman: Y = 1, N = 0				0.020 (0.059)	0.041 (0.057)
Capital city: Y = 1, N = 0					0.255*** (0.043)
Exports (proportion of sales)					0.282** (0.128)
Firm has foreign owners: Y = 1, N = 0					0.679*** (0.075)
Firm purchased fixed assets: Y = 1, N = 0					0.254*** (0.041)
Country fixed effects	No	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	Yes	Yes	Yes
Constant	9.856*** (0.047)	7.755*** (0.181)	7.869*** (0.514)	7.129*** (0.527)	7.216*** (0.528)
Number of observations	42,742	42,742	42,742	42,742	42,742
R-squared	0.014	0.300	0.357	0.364	0.383

Note: Robust standard errors in parentheses. Significance is denoted by ***1% and **5%.
N = no; Y = yes.

burden, and enforcement quality obtained from the ES. For corruption, we use country-level average of the reported bribes paid by the private firms to public officials to “get things done” and expressed as a percentage of the firms’ annual sales (corruption). For the regulatory burden, we use the proportion of firms in the country that were inspected or required to meet with tax officials during the past year (inspections). For the enforcement of rules and laws, we use country-level average of scores reported by the firms on the ES question on whether the functioning of courts is no obstacle (0), minor (1), moderate (2), major (3), or very severe (4) obstacle to their operations (courts obstacle). Note that higher values of courts variable imply weaker enforcement.

5.2 | Base regression results

The baseline regression estimates the linear relationship between labor productivity and informal competition, and results are provided in Table 4. These results reveal a large and statistically significant (at 1% level) negative relationship between the labor productivity and informal competition. This holds with or without the various controls and for all the specifications. Without any controls, the estimated coefficient value of the informal competition variable equals -0.698 (column 1). Controlling for country fixed effect causes the estimated coefficient value to decline sharply (in absolute value) from -0.698 above to -0.217 (column 2). This sharp decline suggests that a large part of the negative labor productivity and informal competition relationship comes from across-country rather than within-country factors. Controlling for the various firm characteristics increases (in absolute terms) the estimated coefficient value of informal competition from -0.217 above to -0.262 (column 5). The estimates in Table 4 (with country fixed effects) imply that the average labor productivity of a formal firm that faces the highest level (equal to 1) of informal competition is lower by 19.5%–24% than for a firm that does not face any informal competition. The corresponding figure for the final specification (column 5) is 23%.

Higher exports and having foreign owners are associated with significantly higher labor productivity. Other controls such as age of the firm, manager experience, being in the capital city, and purchase of fixed assets also correlate positively with labor productivity. In contrast, having more workers is significantly associated with lower labor productivity.

5.3 | Interaction term results

Table 5 shows the estimation results with the interaction terms. Starting with the final specification given earlier (column 5, Table 4), we add the interaction term between informal competition and GDP per capita (column 2). Next, we add the interaction terms between informal competition and corruption, inspection, and courts obstacles. These interaction terms are added one by one in separate regressions (columns 2–4) and then all at once (column 5).

Regression results in Table 5 reveal that the relationship between labor productivity and informal competition is strongly heterogeneous. All the interaction terms are large and statistically significant at the 5% or 1% level. The interaction term with GDP per capita is positive, implying that informality hurts labor productivity of the formal sector firms more in the relatively poorer countries. One reason for this could be that the business environment in the relatively poor countries is less conducive for firms to operate formally. This is confirmed by the other interaction term results in columns 2–5, which reveal that the negative relationship between informal competition and labor productivity of the formal firms is significantly larger (more negative) in countries with higher corruption, more

TABLE 5 Labor productivity of formal firms facing informal competition

Dependent variable: labor productivity	(1)	(2)	(3)	(4)	(5)
Informal competition (cell average)	−2.899*** (0.701)	−0.341*** (0.089)	−0.331*** (0.125)	0.586*** (0.187)	0.808*** (0.219)
Number of workers (logs)	−0.206*** (0.021)	−0.240*** (0.022)	−0.239*** (0.022)	−0.236*** (0.022)	−0.239*** (0.021)
Firm's age (logs)	0.178*** (0.030)	0.238*** (0.031)	0.265*** (0.034)	0.223*** (0.033)	0.222*** (0.033)
Top manager experience (logs)	0.132*** (0.036)	0.174*** (0.037)	0.249*** (0.038)	0.209*** (0.038)	0.171*** (0.037)
Top manager is a woman: Y = 1, N = 0	−0.017 (0.069)	0.034 (0.075)	0.053 (0.078)	0.047 (0.077)	0.030 (0.077)
Capital city: Y = 1, N = 0	0.276*** (0.039)	0.035 (0.038)	0.039 (0.038)	0.093** (0.038)	0.078** (0.038)
Exports (proportion of sales)	0.346*** (0.131)	0.363*** (0.144)	0.362*** (0.149)	0.303** (0.148)	0.298** (0.143)
Firm has foreign owners: Y = 1, N = 0	0.851*** (0.073)	0.811*** (0.076)	0.750*** (0.077)	0.803*** (0.076)	0.833*** (0.076)
Firm purchased fixed assets: Y = 1, N = 0	0.319*** (0.043)	0.357*** (0.044)	0.369*** (0.047)	0.359*** (0.045)	0.357*** (0.044)
GDP per capita (PPP, 2009 international dollars)	0.479*** (0.051)				
Informal competition*GDP per capita	0.291*** (0.081)				
Corruption (country-level average)		−0.166*** (0.031)			−0.191*** (0.034)
Informal competition*corruption		−0.234*** (0.049)			−0.111** (0.052)
Courts obstacle (country-level average)			−0.011 (0.093)		0.212** (0.095)
Informal competition*courts obstacle			−0.369*** (0.145)		−0.405*** (0.142)
Inspections (country-level average)				−0.207 (0.209)	−0.052 (0.195)
Informal competition*inspections				−2.215*** (0.350)	−1.523*** (0.335)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Constant	4.157*** (0.593)	7.515*** (0.555)	7.194*** (0.528)	7.137*** (0.607)	7.118*** (0.614)
Number of observations	42,742	42,742	42,742	42,742	42,742
R-squared	0.260	0.212	0.163	0.193	0.226

Note: Robust standard errors in parentheses. Significance is denoted by ***1% and **5%.
N = no; Y = yes.

inspections, and worse-functioning courts. These results hold even when we include all the interaction terms simultaneously in the regressions (column 5), implying that they are independent effects and not just mere proxies of each other.

To gauge the magnitude of heterogeneity, consider an increase in the proportion of formal firms that compete against informal firms from 0 to 1 (unit increase). This increase is associated with a sharp decline (about 41%) in labor productivity in countries where half of the firms are inspected. In contrast, in a country where only a quarter of the firms are inspected, the corresponding change in labor productivity is a small *increase* of about 3% (based on column 4, Table 5).

5.4 | Underreporting of output

Formal firms that face competition from the informal firms may underreport output to reduce the tax burden and other compliance costs. Thus, one may argue that underreporting may partly explain the productivity difference between formal firms that compete against informal firms and those that do not. A few points are noteworthy here.

First, formal firms that underreport sales are also likely to underreport the number of workers. Thus, it is not clear if underreporting increases or reduces the observed productivity difference.

Second, we regress the labor productivity of a given firm on informal competition faced by other firms in the cell. It is unlikely that informal competition faced by other firms in the cell affects underreporting by the firm in question although the possibility cannot be ruled out completely.

Third, we hypothesize that it is more difficult to hide output for firms that export, have foreign owners, are in the capital city, and have declared fixed asset purchases during the past year. Indeed, firms that underreport sales are also less likely to report capital expansion (purchase of fixed assets). Similarly, exporting firms must comply with more rules and obtain additional licenses that makes it more difficult for them to underreport sales. If our results are driven by underreporting of sales, we should find that the negative relationship between informal competition and labor productivity is smaller (less negative) for firms that export, have foreign owners, bought assets during the past year, and are in the capital city. We tested this assumption by interacting all these variables with the informal competition variable and adding the interaction term (one by one) to the final specification (shown in column 5, Table 4). However, none of these interaction terms were found to be significant.

6 | CONCLUSIONS

Despite the employment opportunities provided by the informal sector, there is concern that high levels of informality can undermine aggregate productivity and retard overall development. Aggregate productivity may decline in two ways. First, informal firms tend to be highly unproductive compared to the formal sector. Thus, aggregate productivity falls with rising informality. Second, informal competition may hurt the productivity of the formal sector firms, and so higher informality lowers aggregate productivity.

Using firm-level survey data for a cross section of mostly developing countries, we document that the labor productivity of informal firms is only about a quarter of the labor productivity of the formal firms. Moreover, we show that the competition from informal firms negatively affects the labor productivity of formal firms by about 19%–24%. These findings support the two informality-related concerns raised earlier.

Our analysis goes beyond the overall effect. We find that the productivity gap between formal and informal firms is larger among formal firms that are better endowed with productivity-enhancing factors (e.g., location in the capital city and older firms). Thus, informal firms appear less capable of leveraging productivity-enhancing opportunities than formal firms. We also find that informal competition is more detrimental to the labor productivity of formal firms when the business environment is less conducive to operate formally. This suggests that the “unfair” advantage enjoyed by the informal firms over the formal firms is exacerbated by a business environment that offers fewer benefits or imposes greater costs on the formal firms.

Several issues remain to be explored. First, the factors that cause the productivity gap and the channels through which informal competition affects formal firms’ labor productivity need further exploration. This is important to sharpen policy responses to the challenges faced by formal and informal firms. Second, the impact of informal competition on formal firms can be extended to other performance measures such as sales and employment growth and innovation. Third, the importance of other business environment factors—physical infrastructure availability, financial development, policies toward SMEs, and tax exemptions—in shaping the productivity gap and the impact of informal competition on the productivity of formal firms can be explored. We leave these issues for future research.

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DATA AVAILABILITY STATEMENT

Firm-level data for formal and informal firms mentioned here are available online. These data can be downloaded from www.enterprisesurveys.org.

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ENDNOTES

- ¹ Some papers have examined how the formal–informal divide affects factors other than productivity such as profits, total sales, total value added, capital stock, and use of paid labor. See, for example, McKenzie and Sakho (2010).
- ² The data can be downloaded from www.enterprisesurveys.org.
- ³ For the informal surveys, total sales and employment information is available for a typical month in the past year. For the formal surveys, information on sales is available only on an annual basis. Therefore, we multiply the monthly sales of the informal firms by 12 to obtain the annual sales estimate.
- ⁴ There are 35 industries, 512 locations, and 3 firm-size groups. Not all locations have all the industries and size groups.
- ⁵ The formal and informal surveys that we exploit are based on different questionnaires, and therefore, the overlap in the questions is limited to basic firm characteristics such as age and size.
- ⁶ We experimented by replacing the capital city dummy with a dummy that equals 1 if the firm is in the capital city or in city with more than one million population and 0 otherwise. This did not change the results discussed in this section.

REFERENCES

- Amin, M., & Islam, A. (2015). Are large informal firms more productive than the small informal firms? Evidence from firm-level surveys in Africa. *World Development*, 74, 374–385.
- Amin, M., & Soh, Y. C. (2020). Does greater regulatory burden lead to higher corruption? Evidence using firm-level survey data for developing countries. *The World Bank Economic Review*. <http://dx.doi.org/10.1093/wber/lhaa007>
- Aterido, R., Hallward-Driemeier, M., & Pages, C. (2011). Big constraints to small firms' growth? Business environment and employment growth across firms. *Economic Development and Cultural Change*, 59(3), 609–647.
- Bahk, B. K., & Gort, M. (1993). Decomposing learning by doing in new plants. *Journal of Political Economy*, 101(4), 561–583.
- Bartelsman, E., Haltiwanger, J., & Scarpetta, S. (2013). Cross-country differences in productivity: The role of allocation and selection. *American Economic Review*, 103(1), 305–334.
- Benjamin, N., & Mbaye, A. (2012). The informal sector, productivity, and enforcement in West Africa: A firm-level analysis. *Review of Development Economics*, 16(4), 664–680.
- Bloom, N., & Van Reenen, J. (2007). Measuring and explaining management practices across firms and countries. *Quarterly Journal of Economics*, 122(4), 1351–1408.
- De Mel, S., McKenzie, D., & Woodruff, C. (2010). Who are the microenterprise owners? Evidence from Sri Lanka on Tokman versus De Soto. In *Josh Lerner and Antoinette Schoar edited International differences in entrepreneurship* (pp. 63–87). National Bureau of Economic Research (NBER), University of Chicago Press.
- De Mel, S., McKenzie, D., & Woodruff, C. (2011). *What is the cost of formality? Experimentally estimating the demand for formalization* (Working Paper). University of Warwick, Coventry.
- De Rosa, D., Gooroochurn, N., & Görg, H. (2010). *Corruption and productivity: Firm-level evidence from the BEEPS survey* (World Bank Policy Research Working Paper 5348), The World Bank, Washington, DC.
- Distinguin, I., Rugemintwari, C., & Tacneng, R. (2016). Can informal firms hurt registered SMEs' access to credit? *World Development*, 84, 18–40.
- Djankov, S., & Murrell, P. (2002). Enterprise restructuring in transition: A quantitative survey. *Journal of Economic Literature*, 40(3), 739–792.
- Dollar, D., Hallward-Driemeier, M., & Mengistae, T. (2006). Investment climate and international integration. *World Development*, 34(9), 1498–1516.
- Du Rietz, A., & Henrekson, M. (2000). Testing the female underperformance hypothesis. *Small Business Economics*, 14(1), 1–10.
- Echevin, D., & Murtin, F. (2009). What determines productivity in Senegal? Sectoral disparities and the dual labour market. *Journal of Development Studies*, 45(10), 1707–1730.
- Farrell, D. (2004). The hidden dangers of informal economy. *McKinsey Quarterly*, 3, 27–37.
- Fisman, R., & Svensson, J. (2007). Are corruption and taxation really harmful to growth? Firm level evidence. *Journal of Development Economics*, 83(1), 63–75.
- Gelb, A., Mengistae, T., Ramachandran, V., & Shah, M. K. (2009). To formalize or not to formalize? Comparisons of microenterprise data from Southern and East Africa. *Comparisons of Microenterprise Data from Southern and East Africa*, July 20, 2009.
- Gonzalez, A., & Lamanna, F. (2007). *Who fears competition from informal firms?* (Policy Research Working Paper Series 4316), World Bank, Washington, DC.
- ILO (2018). *Women and Men in the informal economy: A statistical picture*, 3rd ed. Geneva: Author.
- Islam, A. M., Gaddis, I., Palacios-Lopez, A., & Amin, M. (2018). *The labor productivity gap between female and male-managed firms in the formal private sector* (Policy Research Working Paper 8445), World Bank, Washington, D.C.
- Jensen, J. B., Robert, H. M., & Stiroh, K. J. (2001). The impact of vintage and survival on productivity: Evidence from cohorts of U.S. manufacturing plants. *Review of Economics and Statistics*, 83(2), 323–332.
- Jovanovic, B. (1982). Selection and evolution of industry. *Econometrica*, 50(3), 649–670.
- La Porta, R., & Shleifer, A. (2008). The unofficial economy and economic development. *Brookings Papers on Economic Activity*, Fall, 275–352.
- La Porta, R., & Shleifer, A. (2014). Informality and development. *Journal of Economic Perspectives*, 28(3), 109–126.
- Loayza, N. (1996). The economics of the informal sector. A simple model and some empirical evidence from Latin America. *Carnegie-Rochester Conference Series on Public Policy*, 45, 129–162.

- Maloney, W. F. (2004). Informality revisited. *World Development*, 32(7), 1159–1178.
- McKenzie, D., & Sakho, Y. S. (2010). Does it pay firms to register for taxes? The impact of formality on firm profitability. *Journal of Development Economics*, 91(1), 15–24.
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6), 1695–1725.
- Mendi, P., & Costamagna, R. (2017). Managing innovation under competitive pressure from informal producers. *Technological Forecasting and Social Change*, 114, 192–202.
- Perry, G. E., Maloney, W. F., Arias, O. S., Fajnzylber, P., Mason, A. D., & Saavedra-Chanduvi, J. (2007). *Informality: Exit and exclusion*. Washington, DC: World Bank.
- Pisani, M. J. (2015). Does informality impact formal sector firms? A case study from Nicaragua. *Journal of Developing Areas*, 49(2), 317–334.
- Pounov, C. (2016). Corruption's asymmetric impacts on firm innovation. *Journal of Development Economics*, 118, 216–231.
- Rosenthal, S. S., & Strange, W. C. (2004). Chapter 49—Evidence on the nature and source of agglomeration economies. *Handbook of Regional and Urban Economics*, 4, 2119–2171.
- Rothenberg, A.D., Gaduh, A., Burger, N.E., Chazali, C., Tjandraningsih, I., Radikun, R., ...Weilant, S. (2016). Rethinking Indonesia's informal sector. *World Development*, 80, 96–113.
- Rozo, S., & Winkler, H. J. (2019). Is Informality Good for Business? The Impacts of IDP Inflows on Formal Firms. *Journal of Human Resources*. <http://dx.doi.org/10.3368/jhr.56.4.1018-9794r2>
- Steel, W. F., & Snodgrass, D. (2008). *World Bank region analysis on the informal economy. Raising productivity and reducing risk of household enterprises diagnostic methodology framework. for WEIGO network and World Bank PREM network*. Draft Sept.
- Syversen, C. (2011). What determines productivity? *Journal of Economic Literature*, 49(2), 326–365.
- Tokman, V. E. (1978). Competition between the informal and formal sectors in retailing: The case of Santiago. *World Development*, 6(9–10), 1187–1198.
- Tybout, J. R. (2000). Manufacturing firms in developing countries: How well do they do, and why? *Journal of Economic Literature*, 38(1), 11–44.
- Wagner, J. (2007). Exports and productivity: A survey of the evidence from firm level data. *The World Economy*, 30(12), 60–82.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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